INFRASTRUCTURE AS CODE / TOIL REDUCTION

AUTOMATE EVERYTHING THAT HURTS

Toil is repetitive, manual, automatable operational work that scales with load rather than with engineering investment. Every manual deployment, manual scaling action, and manual config change is a reliability risk and a productivity drain. If a human must do it repeatedly, it is toil.

KEY BENEFITS

- ✓ Frees engineering time from repetitive tasks for higher-value work
- ✓ Eliminates human error as the cause of configuration and deployment incidents
- ✓ Scales operations without scaling headcount proportionally
- ✓ Creates a reproducible, auditable record of every infrastructure change

REAL-WORLD EXAMPLE

Google's SRE teams operate under a formal 50% toil cap. If operational toil exceeds half of engineering time, the team stops feature work and invests in automation until the cap is restored — enforced policy, not aspiration.

PRACTICAL EXAMPLE

The ops team used to manually provision test environments — 45 minutes per request. They write a Terraform module in one sprint. Now engineers self-serve a full environment in 4 minutes via a Slack command.

REFLECTION QUESTION

What percentage of your team's time last week was spent on repetitive manual operations that could be automated?